WHAT IS ARTIFICIAL INTELLIGENCE

- AI is a branch of computer science that is concerned with automation of intelligent behavior

- AI is about making computers able to perform thinking tasks that humans are capable of.

- AI is concerned with increasing the ability of the computer to think humanly, act humanly, think rationally and act rationally.

Acting humanly

The Turing Test proposed by Alan Turing (1950), was designed to provide a satisfactory operational definition of intelligence. A computer passes the test if a human interrogator, after posing some written questions, cannot tell whether the written responses come from a person or from a computer.

Thinking humanly

The interdisciplinary field of cognitive science brings together computer models from AI and experimental techniques from psychology to construct precise and testable theories of the human mind.

Thinking rationally

Logic
===
A>B , B>C then A>C
===
All metals expand when heated
Iron is a kind of metal

Iron expands when heated

Acting rationally

Robot

A rational agent is one that acts so as to achieve the best outcome or, when there is uncertainty, the best expected outcome.

APPLICATIONS OF AI

Question: What can AI do?

Answer: Everything

Robotic vehicles

- Driverless robotic car named STANLEY
- Desert at 22 mph, finishing the 132-mile course first to win the 2005 DARPA Grand Challenge.
- STANLEY is a Volkswagen outfitted with cameras, radar, and laser rangefinders to sense the environment and onboard software to command the steering, braking, and acceleration.

- **The following year CMU's BOSS won the Urban Challenge, safely driving in traffic through the streets of a closed Air Force base, obeying traffic rules and avoiding pedestrians and other vehicles.**

Speech recognition

United Airlines to book a flight.

Game playing

- IBM's DEEP BLUE
- The value of IBM's stock increased by $18 billion.

Spam fighting

Robotics

- iRobot Corporation
- Roomba robotic vacuum cleaners
- Handle hazardous materials, clear explosives, and identify the location of snipers

Logistics planning

- U.S. forces deployed a Dynamic Analysis and Replanning Tool, DART to do automated logistics planning and scheduling for transportation.
- This involved up to 50,000 vehicles, cargo, and people at a time.
- The AI planning techniques generated in hours a plan that would have taken weeks with older methods.

- **The Defense Advanced Research Project Agency (DARPA) stated that this single application more than paid back DARPA's 30-year investment in AI.**

Machine Translation